Texas A&M University
Department of Electrical and Computer Engineering

ECEN 474/704 – (Analog) VLSI Circuit Design

Spring 2018

Exam #1

Instructor: Sam Palermo

- Please write your name in the space provided below
- Please verify that there are **6 pages** in your exam
- You may use one double-sided page of notes and equations for the exam
- Good Luck!

Problem	Score	Max Score
1		40
2		40
3		20
Total		**100**

Name: _____

UIN: _____

Problem 1 (40 points)

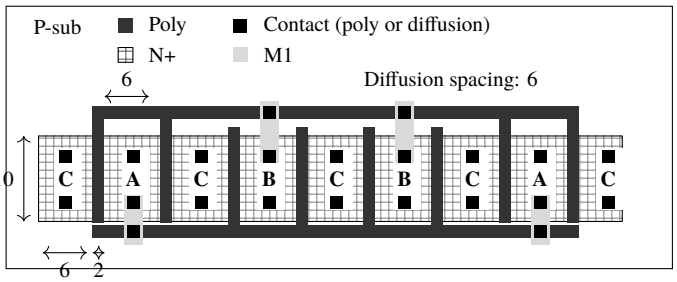
For the layout below, assume that all the commonly labeled diffusion areas are connected with the appropriate metal layers. Assume that $V_A = 2\,\text{V}$, $V_B = 1.5\,\text{V}$, $V_C = 1.0\,\text{V}$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0$, and that all Spice parameters are given (i.e. $C_j, C_{jsw}, C_{jc}, C_{ox}, C_{ov}$). The dimensions are given in μm, with the poly gates having $L_D = 0.1\,\mu$m.

a) Draw the equivalent circuit. Combine all parallel transistors and give the total width and length of the equivalent transistors.

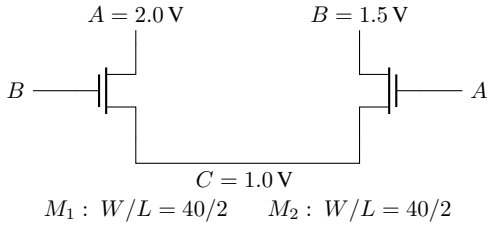
b) What region(s) are the transistors operating in?

c) **For node B only,** give an expression and calculate the total gate cap (only consider where B is the gate).

d) **For node A only,** give an expression and calculate the total junction cap, including the relevant channel-bulk cap. Note for the perimeter terms, include the sides underneath the gate.



Typeset solution. All eight individual transistors have $W/L = 10/2$. Four parallel devices form each equivalent transistor:



$M_1:\quad V_{GS} = 1.5 - 1.0 = 0.5\,\text{V} < V_T \quad \Longrightarrow$ cutoff or subthreshold,

$M_2:\quad V_{GS} = 2.0 - 1.0 = 1.0\,\text{V} > V_T, \quad V_{DS} = 0.5\,\text{V} > V_{GS} - V_T = 0.3\,\text{V} \quad \Longrightarrow$ saturation.

For B, M_1 is cut off. With $L_{\text{eff}} = 2 - 2(0.1) = 1.8\,\mu$m,

$$C_{gB} = C_{gc} + C_{gs,ov} + C_{gd,ov} = (40\,\mu\text{m})(1.8\,\mu\text{m})\frac{C_{ox}C_{jc}}{C_{ox}+C_{jc}} + 2(40\,\mu\text{m})C_{ov},$$

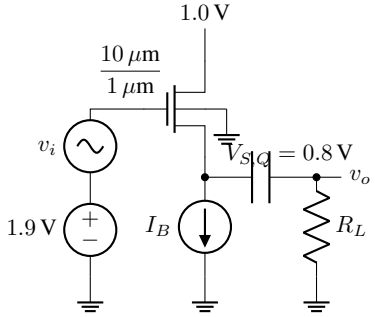
$$\boxed{C_{gB} = 72\,\mu\text{m}^2\frac{C_{ox}C_{jc}}{C_{ox}+C_{jc}} + 80\,\mu\text{m}\,C_{ov}.}$$

For A, the two 10×6 diffusion regions give

$$C_{jA} = A_A C_j + P_A C_{jsw} = 2(10)(6)\,\mu\text{m}^2 C_j + 2\,[2(10+6)]\,\mu\text{m}\,C_{jsw}, \qquad \boxed{C_{jA} = 120\,\mu\text{m}^2 C_j + 64\,\mu\text{m}\,C_{jsw}.}$$

Problem 2 (40 points)

For the circuit shown below, assume the transistor has $V_{T0} = 0.7\,\text{V}$, $\gamma = 0.45\,\text{V}^{1/2}$, and $2\Phi_F = 0.9\,\text{V}$, and its DC drain current is determined by the following equations.



$$I_{DS} = I_o \exp\left(\frac{(V_{GS} - V_{Tn})q}{\zeta kT}\right)\left[1 - \exp\left(-\frac{V_{DS}q}{kT}\right)\right] \quad \text{(Subthreshold)},$$

$$I_{DS} = \mu_n C_{ox}\frac{W}{L - 2L_D}\left(V_{GS} - V_{Tn} - 0.5V_{DS}\right)V_{DS} \quad \text{(Triode)},$$

$$I_{DS} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L - 2L_D}\left(V_{GS} - V_{Tn}\right)^2\left(1 + \lambda V_{DS}\right) \quad \text{(Saturation)},$$

$$V_{Tn} = V_{T0} + \gamma\left(\sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|}\right).$$

a. Calculate V_T and state the transistor's operation region.

$$V_T = 0.7 + 0.45\left(\sqrt{0.9 + 0.8} - \sqrt{0.9}\right) = \boxed{0.860\,\text{V}},$$

$$V_{DS} = 0.2\,\text{V}, \qquad V_{GS} - V_T = 1.1 - 0.86 = 0.24\,\text{V},$$

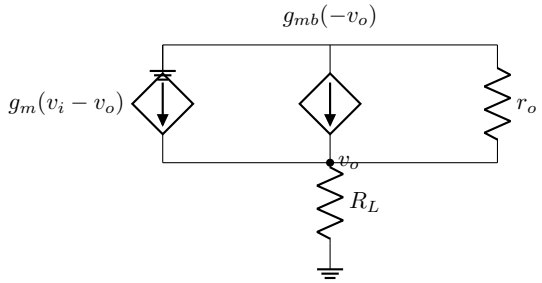
$$V_{DS} < V_{GS} - V_T \implies \boxed{\text{Operation region: Triode.}}$$

b. Using the appropriate I_{DS} equation above, **draw the low-frequency small-signal model of the circuit** (neglecting transistor capacitors), **include expressions for g_m, g_o and g_{mb}, and give an expression for the small-signal voltage gain** $A_v = v_o/v_i$ in terms of the relevant transistor conductances and load resistor. Assume the output AC-coupling capacitor acts as an ideal short.

$$g_m = \left.\frac{\partial I_{DS}}{\partial V_{GS}}\right|_Q = \mu_n C_{ox}\frac{W}{L_{\text{eff}}}V_{DS},$$

$$g_o = \left.\frac{\partial I_{DS}}{\partial V_{DS}}\right|_Q = \mu_n C_{ox}\frac{W}{L_{\text{eff}}}(V_{GS} - V_T - V_{DS}),$$

$$g_{mb} = \left.\frac{\partial I_{DS}}{\partial V_{BS}}\right|_Q = \mu_n C_{ox}\frac{W}{L_{\text{eff}}}V_{DS}\frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}} = \frac{\gamma g_m}{2\sqrt{|2\Phi_F| + V_{SB}}} = \eta g_m.$$



$$v_o = \left[g_m(v_i - v_o) + g_{mb}(-v_o)\right](R_L \parallel r_o),$$

$$v_o\left[1 + (g_m + g_{mb})(R_L \parallel r_o)\right] = v_i g_m(R_L \parallel r_o),$$

$$\boxed{\frac{v_o}{v_i} = \frac{g_m(R_L \parallel r_o)}{1 + (g_m + g_{mb})(R_L \parallel r_o)}}.$$

c. In addition to the parameters given previously, assume $I_o = 1\,\text{nA}$, $kT/q = 25.9\,\text{mV}$, $\zeta = 1$, $\mu_n C_{ox} = 130\,\mu\text{A/V}^2$, $L_D = 0.1\,\mu\text{m}$, and $\lambda = 0.1\,\text{V}^{-1}$. Calculate the necessary load resistor for $|A_v| = 0.7\,\text{V/V}$.

$$A_v = \frac{g_m(R_L \parallel r_o)}{1 + (g_m + g_{mb})(R_L \parallel r_o)} = \frac{g_m}{G_L + g_o + (1+\eta)g_m},$$

$$G_L = \frac{g_m}{A_v} - g_o - (1+\eta)g_m,$$

$$g_m = (130\,\mu)\left(\frac{10}{0.8}\right)(0.2) = 325\,\mu\text{A/V},$$

$$g_o = (130\,\mu)\left(\frac{10}{0.8}\right)(1.1 - 0.86 - 0.2) = 65\,\mu\text{A/V},$$

$$\eta = \frac{0.45}{2\sqrt{0.9 + 0.8}} \simeq 0.173,$$

$$G_L = \frac{325\,\mu}{0.7} - 65\,\mu - 1.173(325\,\mu) \simeq 18.1\,\mu\text{A/V},$$

$$R_L = \frac{1}{G_L} \simeq \boxed{55.4\,\text{k}\Omega}.$$

Rounding note: the supplied solution rounds V_T, g_o, and η before computing G_L. Retaining their full values gives $R_L \simeq 55.83\,\text{k}\Omega$; the displayed $55.4\,\text{k}\Omega$ reproduces the supplied rounded calculation.

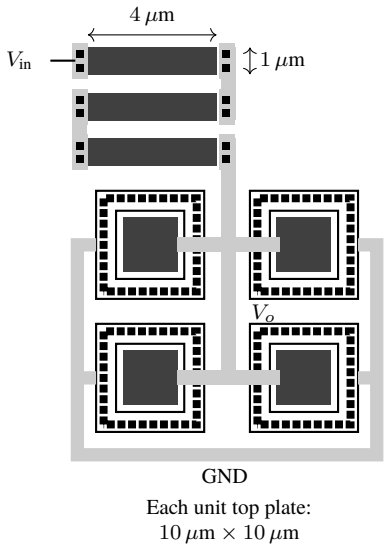
R_L for $|A_v| = 0.7\,\text{V/V} = 55.4\,\text{k}\Omega$

Problem 3 (20 points)

The layout of a simple RC-filter is shown below. For the 3-finger resistor, assume $R_\square = 200\,\Omega/\square$ and each contact has a resistance of $10\,\Omega$. The capacitor consists of 4 unit capacitors that each have top-plate dimensions of $10\,\mu$m $\times$ $10\,\mu$m. These unit capacitors have a capacitive density of $1\,\text{fF}/\mu\text{m}^2$.

a. Sketch the equivalent circuit and give the values of the resistor and capacitor. The resistor contacts should be included in its resistance calculation, but the capacitor contacts' resistance may be ignored. Neglect the impedance of all metal interconnect between the elements.

b. Sketch the Bode plot (magnitude only), and label any pole and/or zero frequencies.



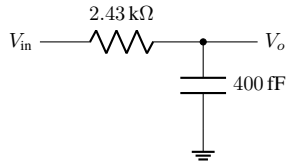
Typeset solution.

n is the number of parallel contacts at each end; F is the number of resistor fingers.

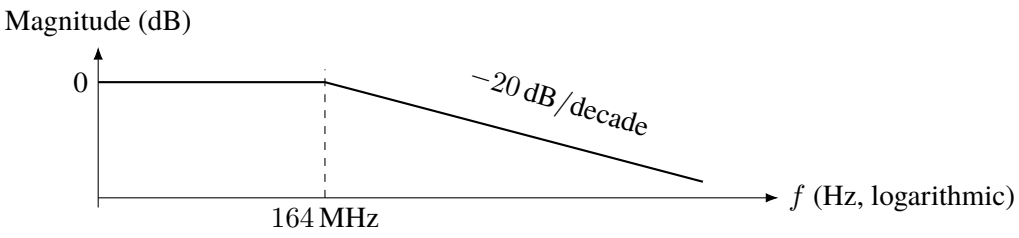
$$\begin{aligned} R_{\text{total}} &= \left(\frac{2R_c}{n} + R_\square \frac{L}{W}\right) F \\ &= \left[\frac{2(10)}{2} + 200\left(\frac{4}{1}\right)\right] 3 \\ &= \boxed{2.43\,\text{k}\Omega}. \end{aligned}$$

With $U = 4$ unit capacitors,

$$\begin{aligned} C_{\text{total}} &= A_{\text{top}} C_{\text{density}} U \\ &= (10\,\mu\text{m})(10\,\mu\text{m})\left(1\,\frac{\text{fF}}{\mu\text{m}^2}\right) 4 \\ &= \boxed{400\,\text{fF}}. \end{aligned}$$



$$\begin{aligned} \frac{V_o}{V_i} &= \frac{1}{1+sRC} = \frac{1}{1+s(9.72\times10^{-10}\,\text{s})}, \\ A_{\text{DC}} &= 1 = 0\,\text{dB}, \\ \omega_p &= -\frac{1}{RC} \simeq -1.03\times10^9\,\text{rad/s}, \\ f_p &= \frac{1}{2\pi RC} \simeq \boxed{164\,\text{MHz}}. \end{aligned}$$



One pole; no finite zero

Scratch Paper